

Part D (Essay Questions)

Answer any two questions (34-36).

Each question carries 10 marks.

34. Trace the curve $(x^2 + y^2)x = a(x^2 - y^2)$, $a > 0$.

35. State and prove Mean Value Theorem.

36. State and prove Fundamental Theorem of Calculus (Part 1).

(2 × 10 = 20 marks)

**FIRST SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

(2016—2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

(Multiple Choice Questions for SDE Candidates)

1. If A and B are two sets such that $A \subseteq B$, then $A \cup B$ is :

- (A) A . (B) B .
(C) $\emptyset$. (D) $A \cap B$.

2. For any two sets A and B , $A - B = \text{—————}$.

- (A) $B - A$. (B) $A \cap \bar{B}$.
(C) $\bar{A} \cap B$. (D) $\bar{A} \cap \bar{B}$.

3. For any two sets A and B , $A - B$ defined by :

- (A) $\{x : x \in A \text{ and } x \in B\}$. (B) $\{x : x \in A \text{ and } x \notin B\}$.
(C) $\{x : x \notin A \text{ and } x \in B\}$. (D) $\{x : x \in A \text{ or } x \in B\}$.

4. The number of subsets of the set $A = \{x : x \text{ is a day of the week}\}$ is :

- (A) 7. (B) 2^6 .
(C) 2^7 . (D) 14.

5. If $A \cap B = A$ and $A \cup B = A$, then :

- (A) $A \subseteq B$. (B) $B \subseteq A$.
(C) $A = B$. (D) None of these.

6. A, B, C are three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$ then :

- (A) $A = B$. (B) $B = C$.
(C) $A = C$. (D) $A = B = C$.

7. If $(2x, x + y) = (8, 6)$ then $y = \text{—————}$.

- (A) 4. (B) 2.
(C) -2. (D) 5.

8. Let $A = \{a, b, c\}$ then the range of the relation $R = \{(a, b), (a, c), (b, c)\}$ defined on A is :

- (A) $\{a, b\}$. (B) $\{c\}$.
(C) $\{a, b, c\}$. (D) $\{b, c\}$.

9. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (1, 2), (2, 1), (2, 3)\}$ be a relation on A . Then R is :

- (A) Reflexive. (B) Symmetric.
(C) Antisymmetric. (D) None of these.

10. If $f(x) = x^2 - 3x + 1$ and $f(2\alpha) = 2f(\alpha)$, then $\alpha =$ _____.

- (A) 3. (B) $\frac{1}{\sqrt{3}}$.
(C) $\frac{1}{\sqrt{2}}$ or $\frac{-1}{\sqrt{2}}$. (D) None of these.

11. Which of the following is not a Proposition ?

- (A) Toronto is the Capital of India. (B) $1 + 1 = 2$.
(C) $x + y = z$. (D) You pass the course.

12. $p \rightarrow q$ is false when :

- (A) p is true and q is true. (B) p is true and q is false.
(C) p is false and q is true. (D) p is false and q is false.

13. Determine which of these conditional statements is false :

- (A) If $1 + 1 = 2$ then $2 + 2 = 5$.
(B) If $1 + 1 = 3$ then $2 + 2 = 4$.
(C) If $1 + 1 = 3$ then $2 + 2 = 5$.
(D) If monkeys can fly, then $1 + 1 = 3$.

14. Let $Q(x, y)$ denote the statement " $x = y + z$ ". Then $Q(3, 0)$ is :

- (A) $3 = 0$. (B) $0 = 3 + 3$.
(C) $3 = 0 + 3$. (D) $3 = 3 + 0$.

15. Which of the following is not an expression for $p \rightarrow q$:
- (A) If p , q . (B) q when p .
(C) p follows from q . (D) p implies q .
16. The bitwise AND of 01 and 11 is :
- (A) 01. (B) 11.
(C) 10. (D) 00.
17. The Compound Propositions p and q are called logically equivalent if $p \leftrightarrow q$:
- (A) A tautology. (B) A contradiction.
(C) A contingency. (D) None of these.
18. The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the true inference called _____.
- (A) Law of detachment. (B) Implication.
(C) Conjunction. (D) Resolution.
19. The solutions of the equation $x^2 + y^2 = z^2$, where x, y, z are integers are called :
- (A) Pythagorean triples. (B) Fermat triples.
(C) Perfect squares. (D) Fermat squares.
20. An invalid argument form often used incorrectly as a rule of inference is :
- (A) Proof. (B) Conjecture.
(C) Theorem. (D) Fallacy.

**FIRST SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

(2016—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Section A

Answer all the twelve questions.

Each question carries 1 mark.

1. Fill in the blanks : If $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{5, 6, 7, 8, 9\}$ are any two sets, $A \oplus B = \underline{\hspace{2cm}}$.
2. Find $n(A \setminus B)$, if $n(A) = 35$ and $n(A \cap B) = 15$.
3. Define an anti-symmetric relation.
4. Give an example of a ternary relation on $\mathbb{R}$, the set of real numbers.
5. Number of reflexive relations on a set of 3 elements is $\underline{\hspace{2cm}}$.
6. Find the domain of the function $f(x) = \sqrt{25 - x^2}$.
7. Define, using indexed collection of sets, the union of arbitrary number of sets.
8. If $\sqrt{25 - x^2} \leq f(x) \leq \sqrt{25 + x^2}$, find $\lim_{x \rightarrow 0} f(x)$.
9. Solve for x and y if $(2x, x - y) = (6, 2)$.
10. The contrapositive of the statement "If I take an umbrella, then it will rain" is $\underline{\hspace{2cm}}$.
11. What can you say about $\lim_{x \rightarrow 0^+} \frac{1}{x}$.
12. Fill in the blanks : The dual of the set equation $(A \cap U) \cup (B \cap A) = A$ is $\underline{\hspace{2cm}}$.

(12 × 1 = 12 marks)

Turn over

Section B

Answer any **nine** out of twelve questions.

Each question carries 2 marks.

13. Find the matrix of the relation R from $A = \{a, e, i, o, u\}$ to $B = \{1, 100, 1000\}$ given by $R = \{(a, 1), (a, 100), (e, 1000), (o, 1)\}$.
14. Evaluate the $\lim_{x \rightarrow 0} f(x)$, if $f(x) = \frac{|x| + x}{2x - |x + 1|^2}$.
15. Construct a relation f on $B = \{a, b, c\}$ which is neither symmetric and nor anti-symmetric.
16. Find the function obtained by shifting the graph of $f(x) = |x|$ right by 2 units.
17. Test whether the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ is injective or not.
18. Define countable set and give an example.
19. Find all real values of x at which $f(x) = \cot x$ is discontinuous.
20. Find the power set of $A = \{-1, 0, 1\}$.
21. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.
22. State the distributive law of disjunction over conjunction.
23. Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers $R^2 = R \circ R$.
24. Show that the set inclusion relation is a partial order on the power set of a set S .

(9 × 2 = 18 marks)

Section C

Answer any **six** out of nine questions.

Each question carries 5 marks.

25. Find f^{-1} by testing its existence when $f(x) = \frac{2x - 3}{5x - 7}$.
26. Find $g \circ f$ and $f \circ g$, if $f(x) = 2x + 1$ and $g(x) = x^2 - 2$.
27. Show that the interval $[0, 1]$ is uncountable.

28. Discuss the continuity of the function $x \sin(1/x)$ at the origin.
29. If $A = \{[-n, n] : n \in \mathbb{Z}\}$, find $\bigcup_{n \in \mathbb{Z}} [-n, n]$ and $\bigcap_{n \in \mathbb{Z}} [-n, n]$.
30. Draw the graph of the function obtained by shifting the graph of $f(x) = x^2 + 1$ down by one unit.
31. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. If $g \circ f$ is one-one, show that f is one-one.
32. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.
33. Translate into English the statement : $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$, where the domain for both variables consists of all real numbers.

(6 × 5 = 30 marks)

Section D

Answer any **two** out of three questions.

Each question carries 10 marks.

34. (a) Show that the relation R on the set of integers given by aRb if $a - b$ is a multiple of 10.
- (b) Find the composition $R \circ S$ and the corresponding matrix if $R = \{(1, 2), (1, 3), (4, 5)\}$ and $S = \{(3, 1), (5, 4), (2, 1)\}$.
35. (a) Find the continuous extension of the function $g(x) = \frac{x^2 - 1}{x - 1}, x \neq 1$.
- (b) Illustrate by examples the distinction between contrapositive and converse.
36. (a) Find $\lim_{x \rightarrow 16} \frac{x^4 - 16}{x - 2}$.
- (b) Show that countable union of countable sets is countable.

(2 × 10 = 20 marks)